윈도우 실행파일 구조(PE파일)-2

3. 32bit, 64bit PE 파일 헤더 분석

이제 실제 32bit, 64bit PE파일을 통해 위에서 살펴보았던 것들을 위주로 분석해보고자 한다. 참고로 Body부분은 생략하고 Header부분의 위주로 분석한다.

3.1 분석 환경

분석환경은 다음과 같다.

1. OS : Windows10(x64)

2. tools : PEView(32bit PE파일), CFF Explorer(64bit PE파일)

3. 대상 : HxD.exe(32bit PE파일), putty.exe(64bit PE파일)

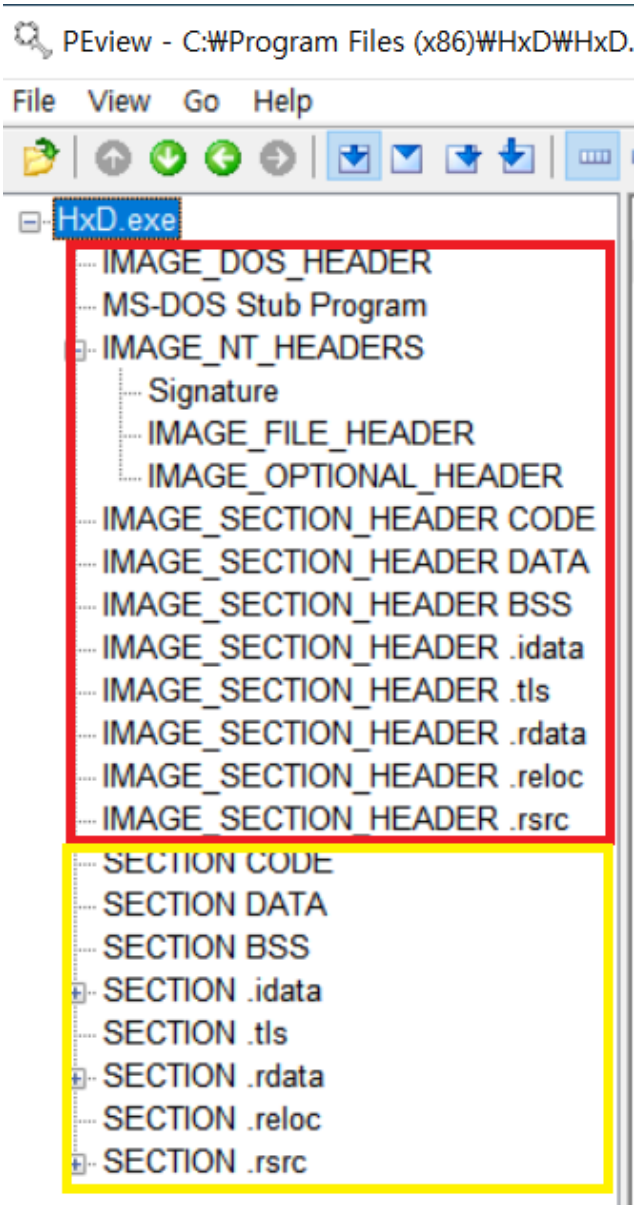
3.2 분석

1) PE View를 통한 HxD.exe분석(32bit PE파일)

확인)

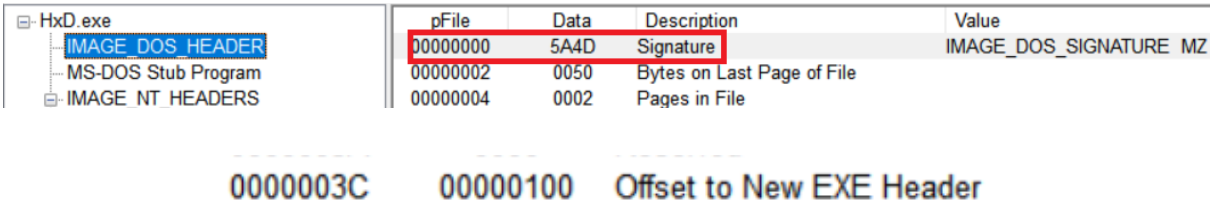
```
▶ file HxD.exe
HxD.exe: PE32 executable (GUI) Intel 80386, for MS Windows
```

구조)



- 빨간색 박스는 PE파일의 Header, 노란색 박스는 Body

1-1) IMAGE_DOS_HEADER(DOS Header)



- PE파일을 나타내는 Signature(e_magic)의 값이 5A4D로 세팅
- NT Header가 시작되는 위치의 옵셋을 나타내는 Offset to New EXE Header(e_lfanew)의 값이 0x100으로 세팅

1-2) MS-DOS Stub Program(DOS Stub)

pFile	Raw Data	Value
00000040	BA 10 00 0E 1F B4 09 CD 21 B8 01 4C CD 21 90 90	. . . . . ! . . L . ! . .
00000050	54 68 69 73 20 70 72 6F 67 72 61 6D 20 6D 75 73	T h i s p r o g r a m m u s
00000060	74 20 62 65 20 72 75 6E 20 75 6E 64 65 72 20 57	t b e r u n u n d e r W
00000070	69 6E 33 32 0D 0A 24 37 00 00 00 00 00 00 00 00	i n 3 2 . . $ 7 . . . . .
00000080	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	. . . . .
00000090	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	. . . . .
000000A0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	. . . . .
000000B0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	. . . . .
000000C0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	. . . . .
000000D0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	. . . . .
000000E0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	. . . . .
000000F0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	. . . . .

- "This program must be run under Win32"라는 문자열이 존재

1-3) IMAGE_NT_HEADERS(NT Header)

- HxD.exe
  - IMAGE_DOS_HEADER
  - MS-DOS Stub Program
  - IMAGE_NT_HEADERS
    - Signature
    - IMAGE_FILE_HEADER
    - IMAGE_OPTIONAL_HEADER
  - IMAGE_SECTION_HEADER CODE

pFile	Raw Data	Value
00000100	50 45 00 00 4C 01 08 00 19 5E 42 2A 00 00 00 00	P E . . L . . . . ^ B * . . . .
00000110	00 00 00 00 E0 00 8F 81 0B 01 02 19 00 26 16 00	. . . . . & . .
00000120	00 D8 04 00 00 00 00 00 CC 33 16 00 00 10 00 00	. . . . . 3 . . . . .
00000130	00 40 16 00 00 00 40 00 00 10 00 00 00 02 00 00	. @ . . . . @ . . . . .
00000140	04 00 00 00 00 00 00 00 04 00 00 00 00 00 00 00	. . . . .
00000150	00 B0 1B 00 00 04 00 00 DC 4B 1A 00 02 00 00 00	. . . . . K . . . . .
00000160	00 00 10 00 00 40 00 00 00 00 10 00 00 10 00 00	. . . . . @ . . . . .

- IMAGE_DOS_HEADER의 멤버 중 NT Header가 시작되는 위치의 옵셋을 나타내는 멤버의 값이 0x100이라 세팅되어 있는대로 IMAGE_NT_HEADERS의 시작 위치 세팅(빨간색 박스)
- 총 3개의 멤버 변수 확인(노란색 박스)

[Signature]

- HxD.exe
  - IMAGE_DOS_HEADER
  - MS-DOS Stub Program
  - IMAGE_NT_HEADERS
    - Signature

pFile	Data	Description	Value
00000100	00004550	Signature	IMAGE_NT_SIGNATURE PE

- Signature값 50450000h("PE"00)확인(변경불가)

[IMAGE_FILE_HEADER]

- HxD.exe
  - IMAGE_DOS_HEADER
  - MS-DOS Stub Program
  - IMAGE_NT_HEADERS
    - Signature
    - IMAGE_FILE_HEADER
    - IMAGE_OPTIONAL_HEADER
  - IMAGE_SECTION_HEADER CODE
  - IMAGE_SECTION_HEADER DATA
  - IMAGE_SECTION_HEADER BSS
  - IMAGE_SECTION_HEADER .idata
  - IMAGE_SECTION_HEADER .tls
  - IMAGE_SECTION_HEADER .rdata
  - IMAGE_SECTION_HEADER .reloc
  - IMAGE_SECTION_HEADER .rsrc

pFile	Data	Description	Value
00000104	014C	Machine	IMAGE_FILE_MACHINE_I386
00000106	0008	Number of Sections	
00000108	2A425E19	Time Date Stamp	1992/06/19 22:22:17 UTC
0000010C	00000000	Pointer to Symbol Table	
00000110	00000000	Number of Symbols	
00000114	00E0	Size of Optional Header	
00000116	818F	Characteristics	
		0001	IMAGE_FILE_RELOCS_STRIPPED
		0002	IMAGE_FILE_EXECUTABLE_IMAGE
		0004	IMAGE_FILE_LINE_NUMS_STRIPPED
		0008	IMAGE_FILE_LOCAL_SYMS_STRIPPE
		0080	IMAGE_FILE_BYTES_REVERSED_LO
		0100	IMAGE_FILE_32BIT_MACHINE
		8000	IMAGE_FILE_BYTES_REVERSED_HI

- **Machine** : 컴퓨터 아키텍쳐 유형을 뜻함(0x14C → x86시스템)
- **Number of Sections** : 0x8로 세팅되어 있으며 섹션의 개수가 8개

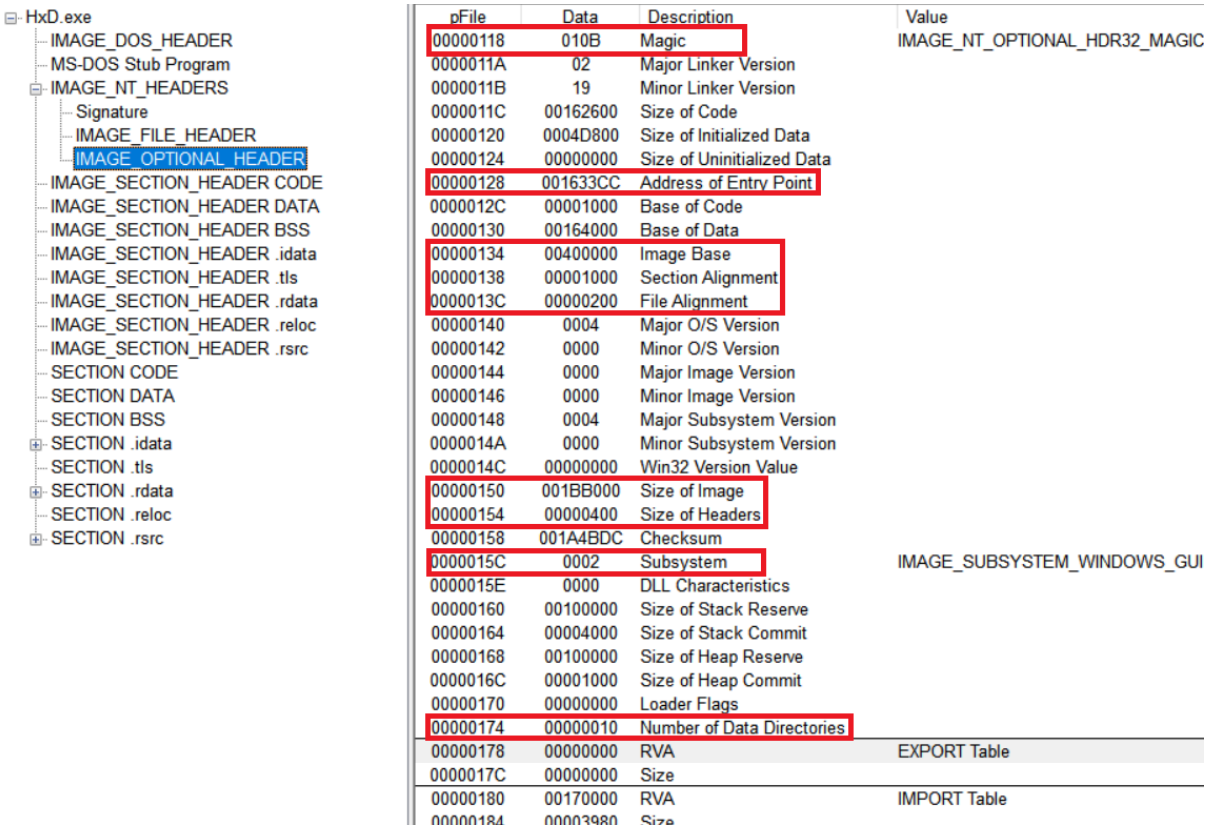
- SECTION CODE
- SECTION DATA
- SECTION BSS
- SECTION .idata
- SECTION .tls
- SECTION .rdata
- SECTION .reloc
- SECTION .rsrc

- 섹션의 개수가 총 8개임을 확인
- **Time Date Stamp :** 이미지를 만든 날짜와 시간을 나타냄(UTC)
- **Size of Optional Header** : Optional Header의 크기가 0xE0

```
00000100  50 45 00 00 4C 01 08 00  19 5E 42 2A 00 00 00 00  PE..L....^B*....
00000110  00 00 00 00 E0 00 8F 81  0B 01 02 19 00 26 16 00  ............&..
00000120  00 D8 04 00 00 00 00 00  CC 33 16 00 00 10 00 00  .........3......
00000130  00 40 16 00 00 00 40 00  00 10 00 00 00 02 00 00  .@....@.........
00000140  04 00 00 00 00 00 00 00  04 00 00 00 00 00 00 00  ................
00000150  00 B0 1B 00 00 04 00 00  DC 4B 1A 00 02 00 00 00  .........K......
00000160  00 00 10 00 00 40 00 00  00 00 10 00 00 10 00 00  .....@..........
00000170  00 00 00 00 10 00 00 00  00 00 00 00 00 00 00 00  ................
00000180  00 00 17 00 80 39 00 00  00 F0 18 00 00 0C 03 00  .....9..........
00000190  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  ................
000001A0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  ................
000001B0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  ................
000001C0  00 50 17 00 18 00 00 00  00 00 00 00 00 00 00 00  .P..............
000001D0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  ................
000001E0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  ................
000001F0  00 00 00 00 00 00 00 00  43 4F 44 45 00 00 00 00  ........CODE....
00000200  A8 24 16 00 00 10 00 00  00 26 16 00 00 04 00 00  .$.......&......
00000210  00 00 00 00 00 00 00 00  00 00 00 00 20 00 00 60  ............ ..`
```

- 다음에 오는 Optional Header의 크기가 0xE0임을 확인
- Characteristics : 파일의 속성을 나타내는 값으로 bit OR연산을 통해 최종적으로 0x818F값으로 설정된 것을 볼 수 있으며 속성 값 중에 0x0002가 존재함을 통해 exe 파일임을 확인

[IMAGE_OPTIONAL_HEADER]



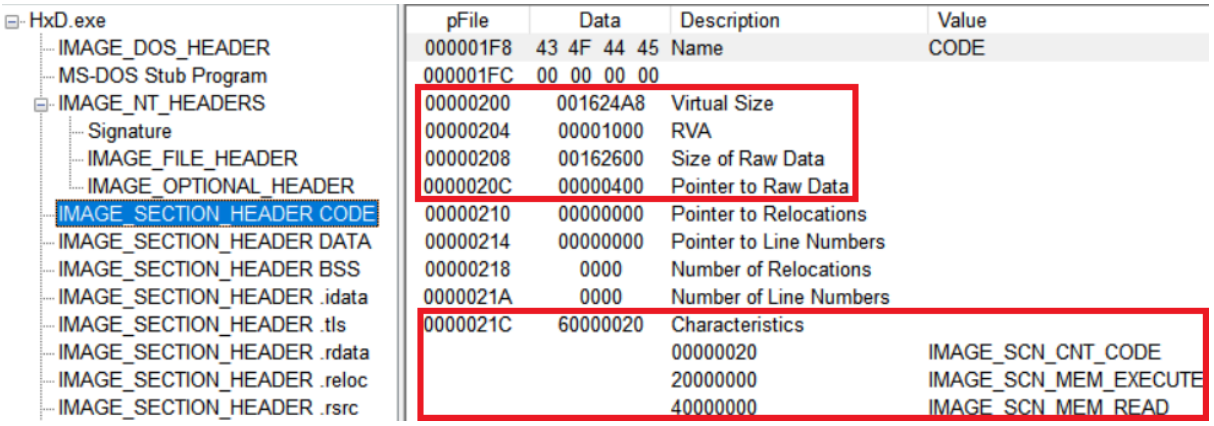
pFile	Data	Description	Value
00000118	010B	Magic	IMAGE_NT_OPTIONAL_HDR32_MAGIC
0000011A	02	Major Linker Version	
0000011B	19	Minor Linker Version	
0000011C	00162600	Size of Code	
00000120	0004D800	Size of Initialized Data	
00000124	00000000	Size of Uninitialized Data	
00000128	001633CC	Address of Entry Point	
0000012C	00001000	Base of Code	
00000130	00164000	Base of Data	
00000134	00400000	Image Base	
00000138	00001000	Section Alignment	
0000013C	00000200	File Alignment	
00000140	0004	Major O/S Version	
00000142	0000	Minor O/S Version	
00000144	0000	Major Image Version	
00000146	0000	Minor Image Version	
00000148	0004	Major Subsystem Version	
0000014A	0000	Minor Subsystem Version	
0000014C	00000000	Win32 Version Value	
00000150	001BB000	Size of Image	
00000154	00000400	Size of Headers	
00000158	001A4BDC	Checksum	
0000015C	0002	Subsystem	IMAGE_SUBSYSTEM_WINDOWS_GUI
0000015E	0000	DLL Characteristics	
00000160	00100000	Size of Stack Reserve	
00000164	00004000	Size of Stack Commit	
00000168	00100000	Size of Heap Reserve	
0000016C	00001000	Size of Heap Commit	
00000170	00000000	Loader Flags	
00000174	00000010	Number of Data Directories	
00000178	00000000	RVA	EXPORT Table
0000017C	00000000	Size	
00000180	00170000	RVA	IMPORT Table
00000184	00003980	Size	

- **Magic** : 32bit인 경우 0x10B를 가지며 **IMAGE_OPTIONAL_HEADER32**를 의미
- **Address of Entry Point** : 메모리 매핑된 후 코드 시작 주소는 0x1633CC로 세팅 (ImageBase값에 이 값을 더해 코드 시작 지점 설정)
- **Image Base** : PE파일이 로딩되는 시작 주소이며 일반적으로 EXE파일은 0x400000
- **Section / File Alignment** : 메모리에서 섹션의 최소 단위는 0x1000이며 파일에서 섹션의 최소 단위는 0x200
- **Size of Image / Headers** : 메모리 로딩될 때 가상 메모리에서 PE Image가 차지하는 크기는 0x1BB00이며 PE Header의 전체 크기는 0x400
- **Subsystem** : 동작 환경을 정의하는 것으로 0x0002이므로 GUI
- **Number of Data Directories** : 총 0x10(16)개의 디렉터리를 갖음

00000178	00000000	RVA	EXPORT Table
0000017C	00000000	Size	
00000180	00170000	RVA	IMPORT Table
00000184	00003980	Size	
00000188	0018F000	RVA	RESOURCE Table
0000018C	00030C00	Size	
00000190	00000000	RVA	EXCEPTION Table
00000194	00000000	Size	
00000198	00000000	Offset	CERTIFICATE Table
0000019C	00000000	Size	
000001A0	00000000	RVA	BASE RELOCATION Table
000001A4	00000000	Size	
000001A8	00000000	RVA	DEBUG Directory
000001AC	00000000	Size	
000001B0	00000000	RVA	Architecture Specific Data
000001B4	00000000	Size	
000001B8	00000000	RVA	GLOBAL POINTER Register
000001BC	00000000	Size	
000001C0	00175000	RVA	TLS Table
000001C4	00000018	Size	
000001C8	00000000	RVA	LOAD CONFIGURATION Table
000001CC	00000000	Size	
000001D0	00000000	RVA	BOUND IMPORT Table
000001D4	00000000	Size	
000001D8	00000000	RVA	IMPORT Address Table
000001DC	00000000	Size	
000001E0	00000000	RVA	DELAY IMPORT Descriptors
000001E4	00000000	Size	
000001E8	00000000	RVA	CLI Header
000001EC	00000000	Size	
000001F0	00000000	RVA	
000001F4	00000000	Size	

- **DataDirectory** : 0x10개의 디렉터리를 가지고 있음

1-4) IMAGE_SECTION_HEADER_CODE



- **Virtual Size** : 메모리에서 섹션의 크기는 0x1624A8
- **Virtual Address(RVA)** : 메모리에서 섹션의 시작주소는 0x1000
- **Size of Raw Data** : 파일에서 섹션의 크기는 0x162600
- **Pointer To Raw Data** : 파일에서 섹션의 offset은 0x400
- **Characteristics** : 섹션의 속성, Bit OR연산 결과 0x60000020

2) CFF Explorer를 통한 putty.exe분석(64bit PE파일)

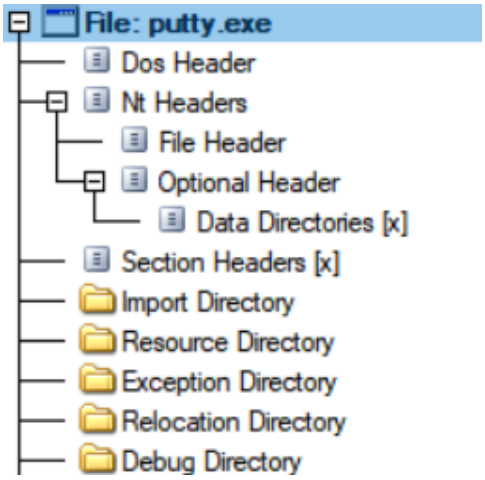
- 위에서 32bit PE파일과 유사하므로 중복되는 내용은 스킵

확인)

```
▶ file putty.exe
putty.exe: PE32+ executable (GUI) x86-64, for MS Windows
```

- 64bit PE파일은 PE32+라고 표현

구조)



2-1) IMAGE_DOS_HEADER(DOS Header)

Member	Offset	Size	Value
e_magic	00000000	Word	5A4D

e_lfanew	0000003C	Dword	00000100

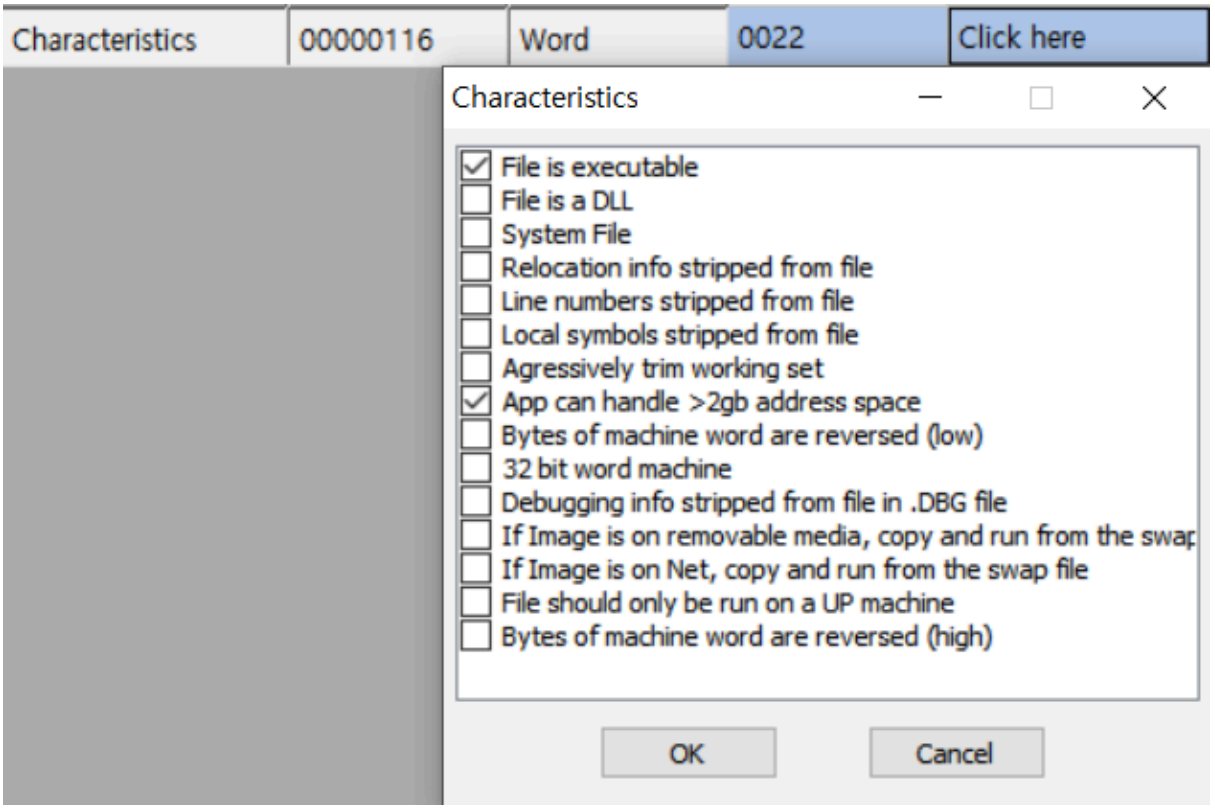
2-2) IMAGE_NT_HEADERS(NT Header)

[Signature]

Member	Offset	Size	Value
Signature	00000100	Dword	00004550

[IMAGE_FILE_HEADER]

Member	Offset	Size	Value	Meaning
Machine	00000104	Word	8664	AMD64 (K8)
NumberOfSections	00000106	Word	0006	
TimeDateStamp	00000108	Dword	58A880D1	
PointerToSymbolT...	0000010C	Dword	00000000	
NumberOfSymbols	00000110	Dword	00000000	
SizeOfOptionalHea...	00000114	Word	00F0	
Characteristics	00000116	Word	0022	Click here



- **Machine** : 컴퓨터 아키텍쳐 유형을 뜻함(0x8664 → x64시스템)

[IMAGE_OPTIONAL_HEADER]

Member	Offset	Size	Value	Meaning
Magic	00000118	Word	020B	PE64
MajorLinkerVersion	0000011A	Byte	0E	
MinorLinkerVersion	0000011B	Byte	00	
SizeOfCode	0000011C	Dword	0008E200	
SizeOfInitializedData	00000120	Dword	0003D200	
SizeOfUninitializedData	00000124	Dword	00000000	
AddressOfEntryPoint	00000128	Dword	0006ED30	.text
BaseOfCode	0000012C	Dword	00001000	
ImageBase	00000130	Qword	0000000140000000	
SectionAlignment	00000138	Dword	00001000	
FileAlignment	0000013C	Dword	00000200	
MajorOperatingSystemVer...	00000140	Word	0006	
MinorOperatingSystemVer...	00000142	Word	0000	
MajorImageVersion	00000144	Word	0000	
MinorImageVersion	00000146	Word	0000	
MajorSubsystemVersion	00000148	Word	0006	
MinorSubsystemVersion	0000014A	Word	0000	
Win32VersionValue	0000014C	Dword	00000000	
SizeOfImage	00000150	Dword	000CE000	
SizeOfHeaders	00000154	Dword	00000400	
CheckSum	00000158	Dword	000CDD88	
Subsystem	0000015C	Word	0002	Windows GUI
DllCharacteristics	0000015E	Word	8160	Click here
SizeOfStackReserve	00000160	Qword	0000000000100000	
SizeOfStackCommit	00000168	Qword	0000000000001000	
SizeOfHeapReserve	00000170	Qword	0000000000100000	
SizeOfHeapCommit	00000178	Qword	0000000000001000	
LoaderFlags	00000180	Dword	00000000	
NumberOfRvaAndSizes	00000184	Dword	00000010	

- **Magic** : 64bit인 경우 0x020B를 가지며 IMAGE_OPTIONAL_HEADER64를 의미